

How Agents Remember the Relationship: Evaluating Relational Memory in Long-Term Dialogue Agents

Anonymous submission

Abstract

As AI agents move from one-shot tools to long-term collaborators, their memory must support more than factual recall. Users may expect an agent to respond in ways shaped by shared events, familiar response norms, evolving user states, and relational boundaries. Inspired by Mutual Theory of Mind, we conceptualize this requirement as relational expectation and study what kind of memory supports it. We introduce ReMem-RE, a controlled evaluation framework for relational memory. ReMem-RE constructs controlled long-term interaction trajectories from sampled personas and recurring event lines, inserts targeted relational probes, and compares a generic memory baseline with three relational memory granularities: conclusion-level, event-summary, and detail-anchor memory. Experiments show that event-summary memory better supports continuity-sensitive and context-specific responses, while detail anchors can improve specificity but also increase memory overuse and boundary risks.

Related Work

Recent work on long-term dialogue memory has increasingly converged on the importance of events and temporal structure. One line of work uses events or sessions as summarization units: LD-Agent describes long-term dialogue memory through event memories, while Reflective Memory Management further summarizes interactions across multiple granularities such as utterances, turns, and sessions (Li et al. 2025; Tan et al. 2025). Another line constructs or evaluates long-context interaction through event-grounded structures: LoCoMo builds very long multi-session conversations from temporal event graphs and evaluates event-related recall and summarization (Maharana et al. 2024). More recent memory frameworks introduce explicitly time-aware or event-centric representations, such as durative memories organized along semantic timelines (Su et al. 2026) and event-like propositions extracted from dialogue sessions for graph-based retrieval (Zhou 2025). Together, these studies show that events are important for long-term memory, retrieval, and temporal grounding. However, these methods still treat events primarily as retrievable content units, summarized episodes, or

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evidence for recall, rather than as the organizing units of a user’s long-term memory. For relationally situated dialogue, this distinction is critical: the agent must recognize not only which past event is relevant, but also which persistent issue the current message belongs to and how that issue has evolved through prior interactions.

Relational Memory Evaluation Framework

We introduce **ReMem-RE (Relational Memory Evaluation for Relational Expectation)**, a controlled framework for evaluating how agents remember the relationship. ReMem-RE is designed around three components. First, it draws on Mutual Theory of Mind (MToM) to operationalize *relational expectation* into evaluation dimensions and targeted relational probes. Second, it constructs controlled long-term interaction trajectories through persona sampling, event-theme sampling, event-line construction, daily interaction generation, and probe insertion, so that relational expectations can emerge over time and be tested in context. Third, it compares agents under different relational memory conditions, allowing us to study whether relationships are better remembered as abstract conclusions, event summaries, or concrete detail anchors. Figure 1 provides an overview of ReMem-RE.

From MToM to Relational Expectation

Mutual Theory of Mind (MToM) provides a useful lens for studying long-term human–AI interaction. Rather than treating interaction as a one-way process in which an agent only infers the user’s immediate intent, MToM emphasizes that humans and agents gradually form expectations about each other’s capabilities, roles, and ways of responding through repeated interaction. From this perspective, a user’s message in a long-term interaction may carry not only a task request, but also an expectation that the agent should respond in a way shaped by their shared interaction history.

We extract from MToM a more operational construct: *relational expectation*. We define relational expectation as the user’s expectation that the agent’s response should reflect prior shared interactions, familiar response norms, evolving user states, and appropriate relational boundaries. A high-quality response is therefore not one that simply recalls more facts, but one that uses prior interaction history to produce a situated, attuned, context-specific, and continuity-sensitive

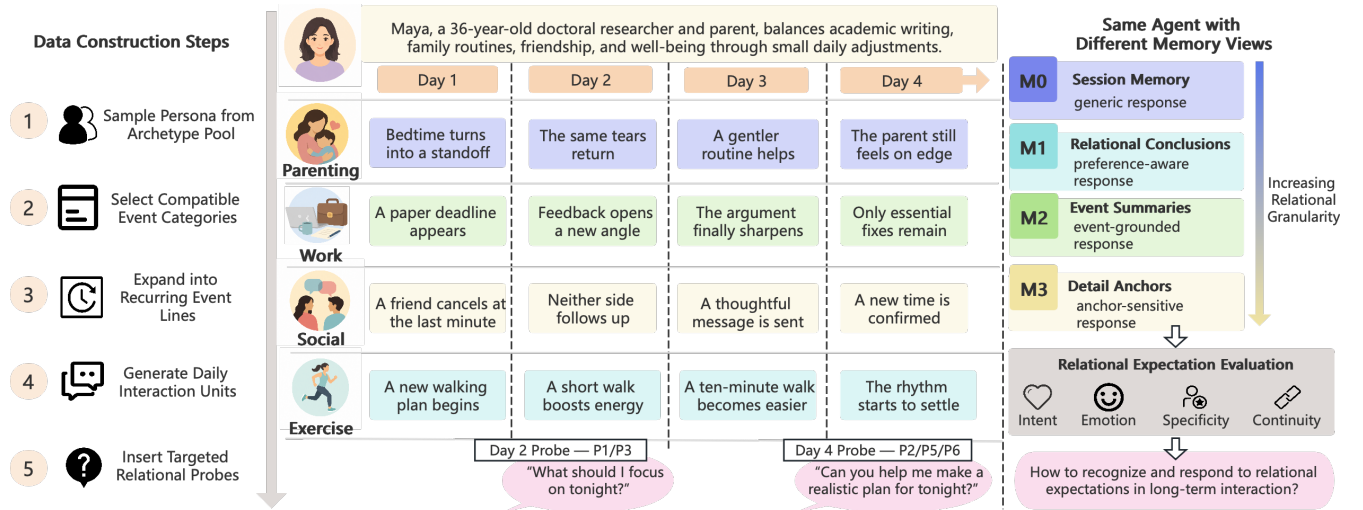


Figure 1: Overview of ReMem-RE. The framework constructs a controlled long-term interaction trajectory with multiple evolving event lines across everyday domains, inserts targeted relational probes at selected time points, and evaluates the same agent under different memory conditions. The goal is not to test factual recall alone, but to examine how different memory granularities support responses to relational expectations formed through shared interaction history.

response. We do not aim to evaluate the overall quality of human-AI relationships, nor do we claim that agents possess genuine Theory of Mind. Instead, we focus on whether an agent can recognize and respond to relational expectations in long-term interaction.

To evaluate whether an agent can recognize and respond to relational expectation, we operationalize it into four high-level evaluation dimensions: D_1 *Situated Intent Understanding*, D_2 *Emotional-State Attunement*, D_3 *Contextual Specificity*, and D_4 *Continuity-Sensitive Response*. These dimensions examine whether the agent understands what the user is really asking for in the situated context, attunes to the user’s current emotional and energy state, grounds its response in the specific interaction context rather than giving generic advice, and responds as a continuous interaction partner without overstepping relational boundaries.

We further design six types of targeted relational probes P_1 – P_6 to elicit and evaluate these dimensions in controlled long-term interaction. Probe types are not intended to be mutually exclusive diagnostic categories. Each probe type has a primary target, but may also involve secondary dimensions because relational expectation is inherently composite. For example, a probe that asks the agent to continue from a previously established handling strategy primarily tests continuity-sensitive response, but it may also reveal whether the response is sufficiently context-specific.

This operationalization allows us to evaluate relational expectation without reducing it to factual recall or generic helpfulness. The next section describes how we construct controlled long-term interaction trajectories in which relational expectations can emerge and be tested under different memory conditions.

Controlled Long-Term Interaction Construction

To evaluate relational expectation, we construct controlled long-term interaction trajectories in which prior events, evolving user states, familiar response norms, and boundary-relevant cues are introduced over time and later tested through targeted probes. Formally, each trajectory is represented as

$$\tau = (z, \mathcal{T}, \mathcal{L}, \mathcal{I}, \mathcal{P}),$$

where z is a sampled user persona, $\mathcal{T}$ is a set of accepted event categories, $\mathcal{L}$ is a set of recurring event lines, $\mathcal{I}$ is a set of daily interaction units, and $\mathcal{P}$ is a set of targeted relational probes. This representation is not intended as a generative model of human relationships, but as a controlled construction framework for long-term interaction contexts.

Archetype-guided persona–event sampling. Prior work has shown the value of synthetic persona construction and temporally evolving user states for evaluating personalization and adaptation (Jiang et al. 2025a,b; Xiao et al. 2025). However, ReMem-RE requires not only plausible personas, but also recurring events that remain coherent with each user’s life situation. We therefore adopt an archetype-guided persona–event sampling strategy. This design is motivated by the need for controlled diversity, persona–event coherence, and cross-sample comparability. Relational expectations do not arise from arbitrary facts attached to a user, but from personally meaningful events that recur and evolve within a plausible life situation. Therefore, the construction process must ensure that the sampled events are compatible with the user’s life stage, social role, resource constraints, and interaction style.

We first sample a persona z from a persona archetype pool, where each archetype specifies a common life situation together with allowable ranges for attributes such as occupation, family structure, life stage, primary life domains, and

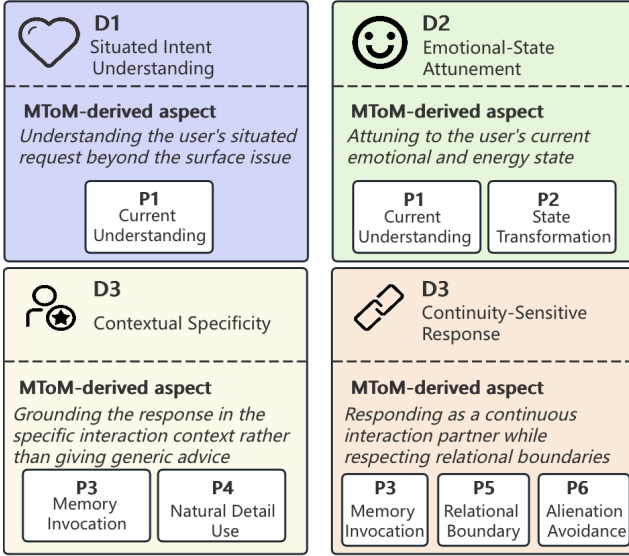


Figure 2: Operationalizing relational expectation into evaluation dimensions and targeted relational probes. We derive four evaluation dimensions from MToM-inspired relational expectation and instantiate them through six targeted probe types. Each probe type primarily targets one dimension but may also reveal secondary aspects because relational expectation is inherently composite.

stress-response patterns. Given z , we then sample candidate event categories from an event-category pool. Each event category represents a persistent life issue or transition that may recur across multiple interaction days. Candidate events are accepted only after persona–event compatibility validation, which enforces both plausibility constraints and domain diversity. The accepted event categories form $\mathcal{T}$, and each of them is expanded into a recurring event line in $\mathcal{L}$, whose stages may include initial concern, recurrence, turning point, partial resolution, pause, or later reflection.

Daily interaction units $\mathcal{I}$ instantiate these event lines into user turns. Each interaction unit contains a scripted opening and a constrained follow-up, generated within a scene boundary so that no new facts, events, or diagnoses are introduced beyond the current scene specification. Finally, targeted relational probes $\mathcal{P}$ are inserted at selected points along the event trajectory. Each probe instantiates one of the predefined six probe types P_1 – P_6 . In this way, probes are embedded into evolving event trajectories rather than presented as isolated questions, allowing us to test whether an agent can respond to relational expectations established through prior interactions.

Memory Conditions and Relational Memory Granularities

Given a controlled long-term interaction trajectory τ , we evaluate the same agent under a set of controlled memory conditions. These conditions are not intended as competing memory architectures, but as memory interfaces that expose different organizations and granularities of long-term mem-

ory. All conditions receive the same user turns and targeted probes; only the available memory interface differs.

M_0 : Session-summary long-term memory baseline. M_0 serves as a strong generic long-term memory baseline. It is motivated by prior long-term dialogue agents that construct long-term memory through session- or time-window-level summarization (Li et al. 2025; Tan et al. 2025). Abstracting from this paradigm, M_0 treats each session or day as the basic unit for long-term memory construction: user messages and M_0 responses are first stored in a short-term session buffer, and each completed session is then compressed into a retrievable long-term session memory. At response time, M_0 retrieves relevant long-term session summaries, ordinary user-profile memories, and conversation snippets, and injects them into the response prompt. M_0 does not access constructed relational memories, event trajectories, relational anchors, probe annotations, or gold response strategies. It therefore tests the limit of generic long-term memory organized as retrievable session summaries, rather than persistent event objects.

While M_0 represents generic long-term memory organized as retrievable session summaries, M_1 – M_3 use an explicit event-aware relational memory construction process. For each interaction window, we identify event candidates, determine whether they create a new event or update an existing one, and maintain a memory unit for each persistent event. Thus, M_1 – M_3 differ not in whether events are detected and updated, but in what level of event-derived information is exposed to the agent: event conclusions, event summaries, or event details.

M_1 : Event-conclusion relational memory. M_1 exposes the relational conclusions distilled from persistent events. These conclusions are directly actionable for response generation: they specify stable response preferences, familiar response norms, boundary constraints, or shared relational conventions. A conclusion may include a concrete phrase or nickname when it has become a stable interactional convention rather than a mere transcript detail. M_1 therefore captures what the agent should remember in order to respond in a familiar and relationship-sensitive way, without exposing the underlying event summaries or detailed evidence from which the conclusion was formed.

M_2 : Event-summary relational memory. M_2 exposes the maintained summaries of persistent event units. Each event memory unit is updated as new interactions occur: when a user message matches an existing event, the event summary is revised to reflect its current status, cross-day progress, user-state changes, unresolved uncertainties, and prior handling strategies; when no existing event matches, a new event unit is created. M_2 therefore allows the agent to understand how an ongoing issue has evolved over time, rather than relying only on distilled relational conclusions.

M_3 : Event-detail relational memory. M_3 further extends M_2 by preserving selected details that ground the event summaries and relational conclusions. These details may include specific scenes, original wording, interaction episodes, or boundary-sensitive cues that explain why a conclusion was

formed or how an event was jointly interpreted. For example, while a familiar nickname or response convention may be stored as an M_1 conclusion, M_3 may preserve the prior interactional scenes or wording that explain why this convention became meaningful. These details are not raw transcript recall; they are event-attached relational anchors used only when they can be naturally and safely invoked.

Although M_1 , M_2 , and M_3 are nested by design, we do not assume that finer-grained memory is always better. The purpose of these controlled conditions is to examine how different levels of event-derived relational memory affect responses to relational expectations.

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